

Locally Plausible, Globally Wasteful: Consensus Consolidation Can Degrade Shared Experience in Multi-Agent LLM Systems

Anonymous submission

Abstract

Multi-agent LLM systems increasingly share experiential memory: agents write distilled insights into a common pool that is retrieved and injected into every teammate’s context. We study a controlled failure mode of one such design that requires no wrong facts, no poisoned text, and no illegal actions. In a trap-maze microscope where every route is legal and shortest-path cost is exactly measurable, we run a preregistered ladder of experiments using implementations based on published components (ExpeL-style insight operations and Generative-Agents-style retrieval). Across a five-arm multi-agent matrix, active memory degrades efficiency and amplifies looping; under our shared-consolidated protocol, the effect is largest relative to the write-only frozen control (success-only excess steps +11.8, 95% CI [7.7, 16.0]; loop rate +0.138, CI [0.092, 0.188]; 5/5 seed signs) and its loop amplification is the only one that significantly exceeds private memory. We do not observe the preregistered read-side importance dose-response; a write-side audit instead finds that consolidation compresses the active strategy supply by $\sim 2.3\times$. The effect persists without self-healing over $T=10$, whereas the exploratory $\lambda=1$ append arm improves over that horizon. Two archive interventions bound, but do not identify, the mechanism: re-exposure at added dose worsens stagnation, and the equal-dose arm fails its preregistered diversity-expansion check. A Qwen3-32B probe replicates compression but leaves the efficiency effect unresolved and excludes a loop amplification of the primary model’s magnitude. These results identify a bounded risk in a measurable setting rather than a task- or model-universal failure.

1 Introduction

When army ants lose the pheromone trail, each ant follows the locally impeccable rule “follow the ant in front of you,” and the colony organizes itself into a circular mill that marches until exhaustion. No single step is wrong; the composition is fatal.

We test whether shared experiential memory in multi-agent LLM systems can admit an analogous failure. The prevailing design pattern—popularized for single agents by experiential learners such as ExpeL (Zhao et al. 2024) and

Reflexion (Shinn et al. 2023), and extended to agent teams by experiential co-learning (Qian et al. 2024a) and cross-task multi-agent experience pools (Li et al. 2025)—lets agents distill natural-language insights from their own trajectories, store them in a memory pool, and retrieve them as guidance. When the pool is *shared* and a reviewer LLM *consolidates* it by merging, editing, and up/down-voting insights, every ingredient is individually reasonable and individually published. We test whether their composition has measurable costs.

The failure we test is *silent*. Agents can keep succeeding, never cross walls, and never submit false answers; the experience text is locally plausible and derived from genuine trajectories. Yet routes can grow longer, agents can revisit the same cells, and loop signatures can rise. Because no error is emitted, success alone need not expose this loss.

Studying such a compositional failure on a live benchmark confounds every knob at once. We instead build a *mechanism microscope*: a generated 15×15 trap-maze family with known shortest paths, local observations, and a step cap, in which efficiency loss and looping are exactly measurable per route (Figure 1). Into this microscope we place concrete implementations of published component families (Table 1). This isolates the tested composition—shared memory plus reviewer consolidation—from maze observability and route-cost ambiguity; it does not make the maze result a substitute for deployment evaluation. All confirmatory experiments were preregistered with frozen gates; every deviation and negative result appears in an append-only log.

Our contributions:

- **A preregistered phenomenon in a controlled setting.** In a five-arm memory matrix over five seeds, the shared-consolidated arm degrades success-only route efficiency, amplifies loops, and reduces success relative to a write-only frozen control. Relative to *private* memory, the shared-specific separation is loop amplification, not a broad efficiency gap.
- **Protocol-level factor separation.** Fresh controls compare joint shared consolidation with per-agent consolidation, fixed-capacity append/agree, read-side MMR, and an exact round-by-round candidate-supply yoke. A pre-specified sensitivity envelope varies retrieval capacity, pool cap, reviewer budget, merge threshold, and recency. Together these separate sharing, consolidation, candidate

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count, and retrieval policy as far as the tested protocols permit, without treating any single contrast as full causal isolation.

- **A bounded mechanism account.** The preregistered read-side importance dose-response is not supported, while a write-side audit shows a $\sim 2.3\times$ compression of active strategy supply. Archive interventions and the new factor controls constrain supply compression as a candidate channel rather than identify it as the cause.
- **Model and task-level boundaries.** A Qwen3-32B probe tests cross-model transfer, and a preregistered six-family BrowserGym MiniWoB study tests executable browser interaction. The latter uses an equal-family, seed-clustered gate with explicit replication, not-detected, and quality-withheld branches rather than assuming transfer from the maze.
- **A reusable microscope and discipline.** The anonymized supplementary artifact contains the environment, runners, prompts, configurations, paired-route statistics, sanitizer, memory audits, and preregistrations with freeze hashes.

2 Related Work

Single-agent experiential learning. ExpeL (Zhao et al. 2024) distills insights from success/failure trajectories with ADD/EDIT/UPVOTE/DOWNVOTE operations and retrieves them as guidance; Reflexion (Shinn et al. 2023) converts failures into verbal self-corrections; ReAct (Yao et al. 2023) interleaves reasoning and acting; Generative Agents (Park et al. 2023) score memories by relevance, importance, and recency; MemoryBank (Zhong et al. 2024) adds long-term memory management. These works established that experience helps a *single* agent. Our write operations, retrieval scoring, and distillation prompts follow their component families (Table 1); we do not claim implementation-identical reproductions of the full original systems.

Multi-agent experience sharing. Experiential co-learning (Qian et al. 2024a) shares shortcut experiences between instructor and assistant agents in ChatDev (Qian et al. 2024b); MAEL (Li et al. 2025) equips each agent in a collaboration graph with an experience pool queried by similarity and reward; multi-agent debate (Du et al. 2024) and self-consistency (Wang et al. 2023) aggregate opinions rather than memories; MetaGPT (Hong et al. 2024) shares a message pool. All published sharing designs accumulate experience additively. To our knowledge, no prior work evaluates the natural next step—*LLM-driven consensus consolidation of a shared experience pool*—with controls; that protocol is exactly our subject.

Memory risk and governance. Recent work studies failure modes in LLM memory systems (Garg et al. 2026) and longitudinal risks of memory-equipped agents (Wang et al. 2026). We complement this line with a preregistered, controlled study of one specific protocol—shared reviewer consolidation—with exact route-level ground truth and explicitly reported negative interventions. For the task-level boundary beyond the maze, MiniWoB from World of

Table 1: Concrete mechanisms adapted from published component families. We retain their operation family or retrieval objective; the shared reviewer update is the evaluated composition, not an implementation-identical reproduction of any full prior system.

Component	Source
Insight ops importance-0 deletion	ADD/EDIT/UP/DOWN, ExpeL
Retrieval scoring	(relevance+importance+recency, λ) Generative Agents
Insight distillation from trajectories	ExpeL
Stepwise solving with injected experience	ReAct-style
Shared pool across agents	Co-learning, MAEL
One reviewer update over a shared pool	composition under test

Bits (Shi et al. 2017), exposed through BrowserGym (Le Sellier De Chezelles et al. 2024), supplies reproducible browser-interaction tasks. We use six families as a controlled transfer boundary, not as evidence about live-web deployment. MemoryArena instead evaluates memory across interdependent multi-session agent tasks (He et al. 2026); our exact-ground-truth microscope trades that breadth for mechanism visibility.

Applicability and governance controls. A complementary line treats memory use as a control decision rather than an unconditional retrieval step. Lu et al. gate memory exposure and selectively accept memory-conditioned answers (Lu et al. 2026); MemArchitect places policy-based adjudication between an agent and its memory store (Kumar, Ba, and Pan 2026); and ProactAgent learns when retrieval is worth invoking during an interaction (Cai et al. 2026). These are not direct baselines for our fixed shared-pool protocol: they add routing, acceptance, or governance policies that our microscope deliberately holds absent. Our contribution is instead a controlled failure characterization that motivates testing such controls against strategy supply, loops, and stagnation, not only task success. RAP likewise retrieves context-matched past experience for planning rather than exposing one globally consolidated pool (Kagaya et al. 2024); this selective design is a natural comparison point for the diversity-aware retrieval tested here.

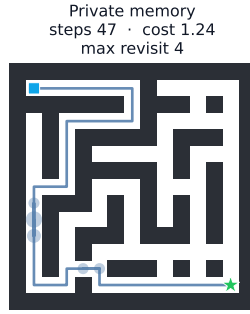
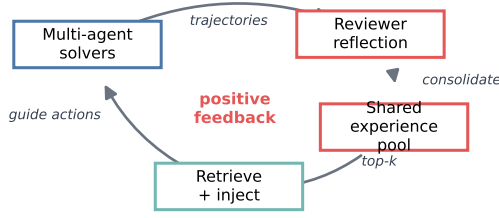
3 The Maze Microscope

Environment. We generate a 15×15 “trap” maze family with enforced shortest-path length ≥ 30 and disjoint train/heldout splits per experiment seed. Solvers receive local observations only (no maze graph, no shortest path), act under a 120-step cap, and are scored by exact excess steps over the known optimum. Efficiency loss and loop signatures are therefore measurable per route—the property that makes silent degradation visible at all.

System under test. Four solver agents attempt the *same* tasks in independent environment copies. They never communicate or vote; the *only* coupling channel is the experience memory. Each round t comprises (i) evaluation on

Figure 1 | Consensus consolidation turns correct shared experience into a silent ant-mill loop

a Shared-experience feedback loop



b Silent degradation on one held-out maze

Same maze, same agent, final round (seed 4). Every route is legal — no wall crossing, no false submit — and every arm still reaches the goal. As the write protocol consolidates experience, the agent revisits more cells and finally loops, while shortest path stays 38 steps.

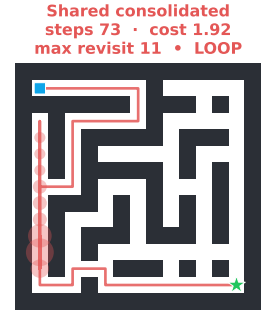
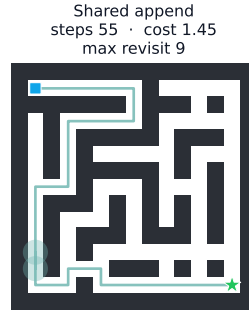


Figure 1: Mechanism microscope and a silent degradation case. (a) The shared-experience feedback loop: solver trajectories are reflected into a shared pool, consolidated by LLM-issued operations, retrieved top- k , and injected back as guidance—a positive-feedback cycle. (b) Same held-out maze and agent, final round; shortest path 38. Revisit heat grows as the write protocol consolidates, culminating in a silent loop under shared-consolidated memory, while every route stays legal and successful.

12 heldout mazes per agent (48 routes) and (ii) a training batch of 4 mazes per agent (16 routes). In the shared-consolidated reference, each training maze then produces one reviewer call over the four agent logs and the shared pool. The private-consolidated control instead invokes the same operation grammar separately for each agent’s single log and private pool: four calls per training maze, each with a six-operation cap, versus one six-operation shared call. This preserves the no-cross-agent-memory intervention but does not equalize reviewer-call or total operation opportunity, so shared versus private is a protocol-level contrast rather than a one-factor causal estimate of sharing alone. The reviewer sees only batched logs and quality signals—never hidden routes. We use “consensus” operationally for the one joint shared reviewer update, not for a solver vote or multi-agent dialogue. The final round evaluates without writing. Retrieval injects the top- $k=6$ insights; the pool is capped at 80.

Protocol detail. In the consolidated arm, the reviewer emits at most six JSON ADD/EDIT/UPVOTE/DOWNVOTE operations over the visible pool. New items begin with two votes; zero-vote items are removed; near-duplicate additions are merged at a fixed similarity threshold. In E2, relevance-only top- k retrieval uses Generative-Agents-style normalization with importance weight $\lambda=0$ and recency weight zero. E3 changes only λ . Full prompts, configurations, sanitizer rules, and retry behavior are included in the anonymized supplementary artifact.

Arms. The five-arm matrix decomposes “multi-agent shared consolidated memory” factor by factor: *no-memory* (is multi-agent execution itself pathological?); *frozen* (reviewer writes, solvers never read: is *storage* harmful?); *private* (per-agent pools: the same operation grammar consolidates each agent’s own pool, but no item crosses agents; reviewer calls are allocated per pool as noted above); *shared-append* (each agent’s distilled reflections enter one shared pool; near-duplicate additions receive a deterministic agreement upvote, but there is no joint reviewer-issued EDIT or DOWNVOTE); and *shared-consolidated* (the protocol under test).

Fresh mechanism and robustness suites. On fresh caches, the five-seed control suite compares shared consolidation with private consolidation, fixed-capacity append/agree (cap 14), and MMR retrieval. MMR greedily reranks GA candidates with $0.70 s_{GA} - 0.30$ times maximum redundancy, where redundancy is cosine similarity between deterministic 128-dimensional token-hashing embeddings; it changes read selection only. A separate append/agree arm is yoked, for every seed and round, to the candidate-pool size produced by shared consolidation; its schedule is extracted without reading behavioral fields. The three-seed one-factor sensitivity suite varies $k \in \{3, 6, 10\}$, pool cap $\in \{40, 80\}$, reviewer budget $\in \{3, 6, 9\}$, merge threshold $\in \{0.70, 0.80, 0.90\}$, and recency weight $\in \{0, 1\}$ without selecting a best setting. Loop rate is the fresh suites’ primary endpoint; the other behavioral and memory endpoints are

prespecified secondary diagnostics.

Browser-task boundary. The MiniWoB study uses six preregistered BrowserGym task families, three seeds, four independent browser environments per task, and frozen, shared-append, and shared-consolidated arms over four rounds. Each arm-seed-family run has 16 training and 12 held-out task instances, a 15-action cap, a pinned MiniWoB++ commit, and a recorded browser and package manifest. After a system-managed Chrome update invalidated an outcome-uninspected partial launch, the complete smoke and formal sequence uses a workspace-local Chrome 150.0.7871.124 runtime pinned by executable and 273-file tree SHA-256 hashes. Formal quality additionally requires exact agreement across all run manifests on the browser runtime, Python and package versions, MiniWoB URL and commit, task registration, and independent-environment count. It also audits failed, retried, and exhausted LLM calls: any terminal route error, or any unresolved held-out, training, or reviewer error on an active-memory path, withholds behavioral interpretation. Nonterminal frozen-arm errors are disclosed separately because that arm never injects its written memory. The consolidated-minus-frozen contrast is primary; append-minus-frozen is secondary, and task families receive equal weight. For MiniWoB, loop/stall burden is the route indicator for either proposing the same normalized action from a previously seen hashed browser state or ending unsuccessfully without an environment termination signal. We report those repeated-action and nontermination components separately. This task-specific boundary analogue is distinct from the maze position-cycle detector below.

Operational loop metrics. For a route $p_0, \dots, p_m$ and goal g , let $d(p, g)$ be Manhattan distance to the goal. We mark a loop if either a position occurs at least ten times, or there is a cycle length $\ell \in \{2, \dots, 8\}$ for which the final 2ℓ positions are two identical consecutive copies and the newest copy reaches no position strictly closer to g than the prefix before it. This detects repeated local behavior without labeling every revisit as a loop. Route-level stagnation is

$$\frac{1}{m} \sum_{i=1}^m \mathbf{1}\{p_i \in \{p_0, \dots, p_{i-1}\}, d(p_i, g) \geq \min_{j < i} d(p_j, g)\}.$$

Thus it counts revisits that fail to improve the best previous goal distance, and is reported separately from loop rate.

Endpoints and statistics. Primary preregistered endpoints: success-only excess steps, loop rate, and the multi-agent ant-mill rate (loop + route overlap). Auxiliary: success, stagnation, maximum revisits, failure-penalized cost, and route diversity. Mechanism audit, logged per run: pool-size trajectory, *distinct injected experiences*, retrieval entropy and top-1 share, sanitizer rejections, and LLM errors. The same-maze same-agent paired bootstrap (10k resamples, 95% CI) describes paired route variation; it does not treat the 240 routes as 240 independent seed replications. For the fresh mechanism, sensitivity, and exact-yoke suites, we instead resample seed clusters and then paired routes within each selected seed, average seed effects equally, and report every seed effect. MiniWoB uses the analogous seed-clustered

Table 2: E2 final-round paired differences versus frozen write-only memory (five seeds, 240 paired routes nested within seeds; success-only excess steps drops unpaired failures). * = paired-route 95% CI excludes zero.

Arm	Succ.	Excess	Loop	Revisit
MAS no memory	−0.004	+1.12	+0.042*	+0.10
Private	−0.175*	+7.32*	+0.079*	+1.20*
Shared append	−0.121*	+2.75	+0.025	+0.23
Shared consol.	−0.171*	+11.84*	+0.138*	+1.68*

bootstrap within each task family before equal-family aggregation. Because excess steps conditions on paired success, we interpret it jointly with success and failure-penalized cost rather than as a stand-alone population efficiency measure. A sanitizer rejects maze IDs, coordinates, and fixed action sequences from memory text; cache keys are salted by (run, seed, round, task, agent, step). The primary model is DeepSeek-V3 (DeepSeek-AI 2024) via a fixed endpoint; a preregistered drift probe (cache disabled, replaying a frozen-arm round; all probe metrics inside the original bootstrap CIs) authorized reuse of cached controls.

4 Preregistered Results: the Phenomenon

4.1 E1: Single-Agent Attribution Control

Before introducing sharing, we ask whether the implemented experience-learning components change single-agent behavior at all. Relative to a no-memory baseline (paired, final round, three seeds), neither the ExpeL-operations writer nor the append writer moves any primary endpoint: success-only excess steps CIs contain zero (append −4.74, CI [−14.1, +4.2]; ops +5.11, CI [−6.4, +16.3]), as do loop rates. By the frozen rule this is a **fail**, recorded as such. However, the mandated adherence analysis shows experience is *not* ignored: the ops writer shifts action selection away from the controller suggestion after memory activates (+0.041 deviation, same sign in 3/3 seeds), and the append writer significantly reduces maximum revisits. The single-agent null is therefore an *attribution control*: it makes a generic effect of merely enabling memory less plausible, but it does not by itself identify sharing as the causal channel.

4.2 E2: Multi-Agent Memory Matrix

We compare the five arms at four agents, $T=6$, five seeds, against the write-only *frozen* control. Table 2 and Figure 2 report paired differences at the final round.

Three readings follow. First, multi-agent execution is not itself pathological: MAS-no-memory matches frozen on efficiency and success, with only a small loop increase (+0.042, CI [0.008, 0.075]); frozen itself matches no-memory, so this comparison does not reveal a storage-only effect. Second, **within this setting, the shared-consolidated arm meets every preregistered degradation criterion**: success-only excess steps +11.84 (CI [7.66, 16.02]), same-sign in 5/5 seeds; loop rate +0.138 (CI [0.092, 0.188]); success −0.171. Third, active injection is a broad risk in this experiment—private

E2: active consolidated memory degrades efficiency and increases loops

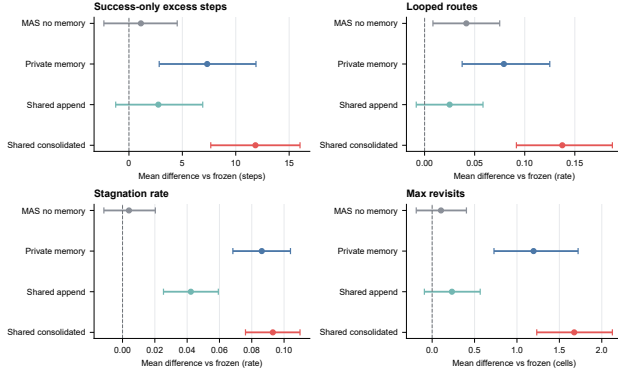


Figure 2: **E2 main result: paired differences versus frozen write-only memory** (five seeds, final round). Points are mean differences; bars are 95% bootstrap CIs; the dashed line is zero. Shared-consolidated (red) degrades on all four end-points with the largest effects; MAS-no-memory (grey) sits on the zero line except for a small loop increase.

memory also degrades—so we test the narrower shared-specific contrast directly.

Shared versus private boundary. The original E2 matrix contrasts the shared arms against a *private* baseline. The efficiency separation is not significant (consolidated vs. private success-only excess steps $+0.87$, CI $[-4.29, +6.15]$), but loop amplification is: consolidated loops significantly more than private ($+0.058$, CI $[0.008, 0.113]$), while append loops *less*. The honest claim is therefore that consensus consolidation specifically amplifies looping relative to private memory; we do not claim a broad shared-over-private efficiency gap.

4.3 E3: Read-Side Dose-Response Is Null

If the harm were read-side—agents increasingly retrieving a few high-vote memories—then raising the importance weight λ in Generative-Agents-style scoring should deepen degradation. We did not observe that preregistered monotone pattern. Sweeping $\lambda \in \{0, 0.5, 1\}$ on the append arm (three seeds, $T=6$), the paired difference of $\lambda=1$ versus $\lambda=0$ on success-only excess steps is $+1.69$ (CI $[-4.26, +7.61]$), loop rate $+0.049$ (CI $[0.000, 0.104]$), and final-round retrieval entropy is non-monotone in λ ($0.833/0.826/0.845$). The read-side usage-weighted-feedback hypothesis is therefore not supported in the tested λ range; this null result is not evidence that every retrieval-side mechanism is harmless.

4.4 Mechanism Audit: Write-Side Strategy-Supply Compression

An offline audit of the same E2 runs locates the concentration on the *write* side (Figure 3). Under identical settings, the consolidated pool converges to 14.2 items with only 11.0 distinct experiences ever injected across all rounds, whereas the append pool grows to the cap (80) and injects 25.4 distinct

experiences, and private injects 37.6. Consensus consolidation compresses the population’s active strategy supply by roughly $2.3\times$ relative to append. The top-1 retrieval share rises in step, but this is a consequence of the smaller pool rather than an independent channel; the load-bearing evidence is the direct pool-size and distinct-injection counts. Given the intervention results below, we report compression as a robust, cross-checked *correlate* and candidate mechanism, not as established cause.

4.5 E4: Long-Horizon Stress Test (Exploratory)

We extend the shared arms to $T=10$ (three seeds; exploratory). Relative to frozen, shared-consolidated remains degraded at $t=3, 6, 9$: success differences are $-0.174/-0.146/-0.257$ and loop differences are $+0.083/+0.083/+0.118$ (all CIs exclude zero). A within-condition late-versus-early comparison detects no further consolidated self-trend, so the gap opens early and persists at a noisy plateau rather than provably deepening. Shared append instead improves over time on cost ratio (-0.079), excess steps (-3.50), stagnation (-0.026), and max revisits (-0.340), with all four CIs excluding zero; this is not a universal append claim. Seed effects remain heterogeneous: seed 1 has the only detected late success decline, seed 2 ends with the worst trajectory, and seed 0 shows no decline.

4.6 The Failure Is Silent

Figure 1(b) shows a case study mined from the runs: the same held-out maze and agent at the final round, whose shortest path is 38 steps. As the write protocol consolidates, the route degrades monotonically—frozen 39 steps (one revisit), private 47 (four), shared-append 55 (nine), shared-consolidated 73 with cost 1.92, eleven revisits to the same cell, and a detected loop—yet every route is legal and every arm still reaches the goal. This is the ant-mill signature: not a false belief, but a locally plausible correction signal, reinforced through consensus writing, that composes into a globally wasteful cycle.

5 Mechanism Interventions and Boundaries

A second preregistration (Phase Gamma) froze two archive interventions and a cross-model probe.

Archive interventions. The additive arm retains discarded variants and injects active top-6 plus archive top-6. Distinct supply roughly doubles, but so does prompt dose. No endpoint improves: success non-inferiority fails (-0.056 , CI $[-0.153, +0.042]$), success-only excess is null (-0.45 , CI $[-6.86, +5.96]$), and stagnation worsens ($+0.045$, CI $[+0.023, +0.067]$). A second arm jointly ranks active and archive candidates under the original total top-6. Its preregistered diversity-expansion check fails ($1.42/1.55/1.07\times$ across seeds), despite archive selection in 896/912 eligible retrievals. We therefore do not evaluate its behavioral gate or claim that equal-dose archive rescue is ineffective. Added-dose re-exposure can backfire, while one fixed-capacity joint ranking does not establish restored diversity; compression remains a correlate with an unresolved causal role.

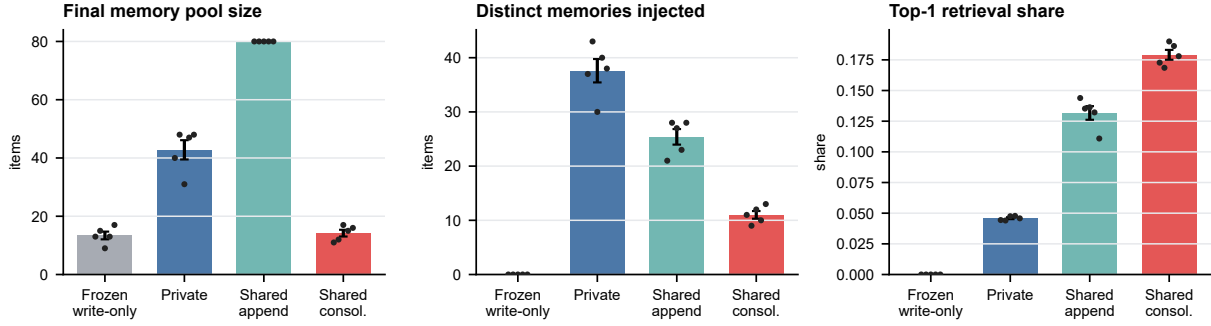


Figure 3: **Mechanism audit: consolidation compresses the strategy supply.** Final pool size, distinct experiences ever injected, and top-1 retrieval share across five seeds (dots). Consensus consolidation keeps a small pool and injects the fewest distinct strategies, concentrating retrieval; the direct counts (left, centre) carry the claim.

5.1 Cross-Model Probe: Compression Replicates, Coupling Is Not Detected

We repeat the frozen/append/consolidated triangle on Qwen3-32B (Qwen Team 2025) (three seeds, $T=4$). The pre-registered gate is *not-detected-or-underpowered*. Efficiency is direction-consistent but unresolved: success-only excess is $+5.83$ (CI $[-3.75, +15.20]$, all three seed signs positive). Loop rate is -0.076 (CI $[-0.153, 0.000]$), excluding the primary model’s $+0.138$ amplification, and success shows no drop ($+0.042$, CI $[-0.049, +0.132]$). Yet consolidation still compresses distinct injected supply by $1.55\times$ on Qwen versus $1.68\times$ on DeepSeek. Similar structural compression with different detected behavior argues against treating compression alone as sufficient cause. The MiniWoB boundary is reserved for the main-text transfer display once its formal evidence is complete; full Qwen diagnostics remain in the artifact.

6 Design Implications

- **Treat shared consolidation as an audited design choice.** In this setting, the same infrastructure degrades under reviewer consolidation while the exploratory $\lambda=1$ append arm improves over a longer horizon. Consolidation should therefore be measured rather than assumed to be hygiene.
- **Monitor the write side in addition to success.** Distinct injected experiences, pool-size trajectory, and retrieval concentration are logged-at-source signals; success alone can miss efficiency decay.
- **Track loops and stagnation in agent telemetry.** Repeated state-action signatures and stagnation separated consolidated from private memory when aggregate efficiency did not.
- **Do not assume an archive fixes consolidation.** Added-dose re-exposure worsened stagnation here, while the equal-dose intervention did not meet its diversity manipulation check. A retained archive must also be retrievable under the deployed ranking rule.

7 Limitations

The maze is the only environment in which shortest-path cost and position cycles are known exactly, and all studies use one primary model, DeepSeek-V3. The six-family MiniWoB boundary adds executable browser interaction but remains a controlled synthetic benchmark rather than an open-web or deployment workflow. The Qwen probe bounds but does not establish cross-model transfer: efficiency is unresolved (underpowered), and the loop signature does not carry at the primary-model magnitude. Strategy-supply compression also remains a candidate channel. The exact-size yoke equalizes candidate count but not item content or write history; the MMR control changes only read selection through deterministic token-hashing cosine; and no deterministic writer-compression baseline is included. Thus the interventions constrain, but do not identify, compression as the cause of any behavioral separation. The fresh maze suites use a seed-clustered hierarchical bootstrap, but the controls and exact yoke have five seeds, the sensitivity suite has three, and MiniWoB has three seeds per task family. Seed-level heterogeneity is therefore reported descriptively rather than treated as many independent replications. Success-only excess steps is conditional on success and must be read jointly with success and failure-penalized cost. Effects may differ with stronger reviewers, other consolidation prompts, or retrieval schemes beyond similarity ranking. The MMR control tests one reproducible lexical diversity policy, not learned semantic MMR, xQuAD, or diversity-preserving retrieval generally. The private-consolidated control also reallocates reviewer calls by pool—four per-agent calls versus one joint shared call per training maze—so it removes cross-agent memory access without isolating sharing from reviewer prompt composition or total operation opportunity.

8 Conclusion

Shared experiential memory warrants measurement as well as capability evaluation. In a preregistered maze microscope, the tested shared reviewer-consolidation protocol can reduce

route efficiency, amplify loops relative to private memory, persist over a longer horizon, and compress the active strategy supply. The archive interventions do not identify compression as the cause: added-dose re-exposure worsens stagnation, while the equal-dose intervention fails its preregistered diversity-expansion check. The Qwen probe shows that compression can transfer without detecting the observed behavioral coupling. The ant-mill is therefore a bounded warning, not a claim of universal degradation: locally plausible experience can compose into globally wasteful behavior.

Reproducibility. The anonymized supplementary artifact contains preregistrations with SHA-256 freeze records, decision logs, per-route artifacts, memory audits, prompts, configurations, statistics code, and maze runners. Every number in this paper maps to a frozen gate report or an explicitly labeled exploratory analysis.

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